

Interval volume from thinned causal orders: finite confidence and calibration limits

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10 October 2026

Version 0.1.0 — unpublished manuscript

Abstract

An event detector samples a causal geometry with an unknown spatially varying retention probability. We study the normalized physical volume of one marked chronological interval when only the full order of the retained events and two distinguished anchors is observed. In a smooth, bounded class of two-dimensional conformal geometries, with known detector ratio bound $1 \leq R \leq 2$, the fixed-confidence minimax accuracy for $n \geq 1$ retained events has scale $\min\{1, n^{-1/2} + (R - 1)/(R + 1)\}$, uniformly in n and R . The lower bound covers arbitrary measurable estimators of the full marked order, including independent randomization. It uses admitted geometric alternatives with either close sampling laws or exactly equal detected laws. We separately give a finite confidence report by composing certified binomial endpoints with the established bounded-selection transforms of Aronow and Lee. Complete thinning arguments connect fixed, independently mixed, Poisson and first- n retention experiments. The mathematical chain has exact-source Lean verification on Google Cloud. A preserved pilot and a separate frozen finite comparison support retaining the exact-binomial report for practical reporting; the minimax upper bound is conservative. No absolute volume scale or physical detector calibration is inferred.

1 Question and contribution

Counting the events between two marked events is an intrinsic operation on a causal order. Its physical interpretation depends on how events are sampled. If detection favors some locations, the proportion of retained events in an interval estimates its *detected* probability, rather than its physical volume fraction. More retained events alone need not remove this discrepancy.

This paper connects three results in one explicit experiment: the law of independent thinning, an executable finite confidence report under bounded unknown detection, and a uniform upper and lower accuracy theorem for a geometric class. The lower construction preserves the entire marked-order law, not just the interval count. Consequently, an estimator cannot avoid the calibration obstruction by using additional order relations or an independent random seed. A second construction gives the sampling term even under perfect detection.

The statistical ingredients have established antecedents. The binary selection transforms are those of Aronow and Lee [1]; confidence inversion follows Clopper and Pearson [2], and the analytic comparison uses Hoeffding’s inequality [3]. Partial identification distinguishes an unknown target from bounds on its possible values, and confidence for either object requires an additional sampling argument [4, 5, 6]. Our contribution is the connected geometric experiment, its admitted full-law

alternatives, the finite uniform rate, and the formal and computational evidence. We make no exhaustive priority claim for all related statistical problems.

The physical question here is narrower than reconstructing a complete metric from order. The target is one normalized interval volume in a specified two-dimensional model. Coordinates define the model but are not observed. The fixed anchors and their roles are part of the experiment; choosing an interval after looking at the sample would require a different coverage argument.

2 Geometry, observations and the main result

Let $D = [0, 1]^2$ and write $x = (u, v)$. A function $\rho : \mathbb{R}^2 \rightarrow \mathbb{R}$ belongs to $\mathcal{K}$ when it is smooth on some open neighborhood of D and, for all points of D , satisfies

$$\begin{aligned} \frac{1}{2} \leq \rho \leq \frac{3}{2}, \quad |\rho(x) - \rho(y)| \leq 2\|x - y\|_2, \\ \int_0^1 \rho(u, t) dt = 1, \quad \int_0^1 \rho(t, v) dt = 1. \end{aligned} \tag{1}$$

The Lipschitz norm is Euclidean. Only behavior on D and smoothness near it are required; positivity on all of $\mathbb{R}^2$ is not assumed. Set $\mu_\rho(B) = \int_{B \cap D} \rho(u, v) du dv$.

Proposition 2.1 (Geometric interpretation). *The measure μ_ρ is a probability measure. On D , equip the model with the Lorentzian metric*

$$g_\rho = -\rho(du \otimes dv + dv \otimes du)$$

and future direction increasing both coordinates. Its volume element is $\rho du dv$ and its strict chronological relation is $x \prec y$ if and only if $x_1 < y_1$ and $x_2 < y_2$. For the fixed anchors $a = (0, 0)$ and $b = (1/2, 1/2)$, let

$$A = \{x : a \prec x \prec b\} = (0, 1/2)^2, \quad v_\rho = \mu_\rho(A).$$

Then $1/8 \leq v_\rho \leq 3/8$.

Proof. Integrating either marginal identity gives total mass one. The metric matrix has determinant $-\rho^2$, so its volume density is ρ . A nonzero future timelike tangent has both components positive, since $g_\rho(\dot{x}, \dot{x}) = -2\rho\dot{u}\dot{v} < 0$ and the time orientation selects the positive branch. A future timelike curve therefore increases both coordinates. Conversely, the straight segment between two points with strictly increasing coordinates lies in the convex diamond and is future timelike. This proves the relation and the expression for A . Its ordinary area is $1/4$; integrating the density bounds gives the asserted range. Boundaries have zero measure. $\square$

Our units normalize the total geometry volume to one. Thus v_ρ is a dimensionless physical volume fraction. Without external scale information this experiment cannot attach an absolute spacetime volume unit to the answer.

Let the detector $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$ be measurable, with the global bounds

$$1/R \leq \pi(x) \leq 1, \quad R \geq 1. \tag{2}$$

Here R is known and deterministic. Write

$$Z = \int \pi d\mu_\rho > 0, \quad \nu(B) = Z^{-1} \int_B \pi d\mu_\rho, \quad \theta = \nu(A).$$

In the fixed-retained-count experiment, $X_1, \dots, X_n$ are independent with law ν . The observation Y_n consists of every directed strict chronological relation among these events and the two anchors, modulo permutations of sample labels fixing the anchor roles. This is a finite measurable observation space. It includes sample–sample, anchor–sample and anchor–anchor relations. Coordinates, rejected locations, detector values, generated counts and clocks are not observed in this experiment.

The count

$$K = \sum_{i=1}^n \mathbf{1}\{a \prec X_i \prec b\}$$

is determined by Y_n and unchanged by relabeling. Under the model, $K \sim \text{Bin}(n, \theta)$. The complete observation can contain more information than K ; only the upper procedure discards that information. Write $P_{\rho, \pi}^{(n)}$ for the law of Y_n .

Theorem 2.2 (Uniform joint accuracy). *For $n \geq 1$ and $1 \leq R \leq 2$, put*

$$a_n = n^{-1/2}, \quad b_R = \frac{R-1}{R+1}, \quad s(n, R) = \min\{1, a_n + b_R\}.$$

For every admitted (ρ, π) , the observable fraction $\hat{v} = K/n$ satisfies

$$P_{\rho, \pi}^{(n)}\{|\hat{v} - v_\rho| > 2s(n, R)\} \leq 1/20. \quad (3)$$

Conversely, let (Ω, ξ) be any probability space and let $T : Y_n \times \Omega \rightarrow \mathbb{R}$ be any measurable estimator. There exists an admitted (ρ, π) such that

$$(P_{\rho, \pi}^{(n)} \otimes \xi)\{|T - v_\rho| > s(n, R)/256\} \geq 1/4. \quad (4)$$

The seed law ξ is common to the alternatives and independent of the observation. No boundedness or determinism restriction is placed on T .

The proof appears in Section 5. If the 95% minimax radius is the infimum over eligible procedures of radii with worst-case failure at most $1/20$, this theorem places it between $s(n, R)/256$ and $2s(n, R)$. It holds at $R = 1$ and uniformly for sequences $R = R_n$ tending to one. The constants are deliberately conservative and are not asserted optimal. Section 4 supplies the more useful reporting procedure. Zero samples are treated separately in Proposition 5.5.

3 What independent thinning produces

We give the process argument on an arbitrary probability space (E, μ) with a measurable $0 \leq \pi \leq 1$ and $Z = \int \pi d\mu > 0$. Generate independent pairs (X_i, U_i) , where $X_i \sim \mu$ and $U_i \sim \text{Unif}[0, 1]$, with all coordinates independent. Retain event i precisely when $U_i < \pi(X_i)$, and list retained locations in increasing *generation index*. This order is not a sorting by spacetime location.

Proposition 3.1 (Finite and mixed generation). *Let $\alpha(B) = \int_B \pi d\mu$ and $\nu = \alpha/Z$. For fixed generated count N , a retained index set $I = \{i_1 < \dots < i_n\}$, and measurable $B_1, \dots, B_n$, the joint probability of exactly this index set and $X_{i_j} \in B_j$ for every j is*

$$Z^n (1 - Z)^{N-n} \prod_{j=1}^n \nu(B_j). \quad (5)$$

Writing C_N for the retained count and V_n for its ordered locations, their joint subprobability measure is

$$\mathbb{P}(C_N = n, V_n \in \cdot) = \binom{N}{n} Z^n (1 - Z)^{N-n} \nu^{\otimes n}(\cdot) \quad (0 \leq n \leq N). \quad (6)$$

If N is chosen independently with any probability mass function q_N , the right side is summed against q_N . Conditional on any retained count of positive probability, the locations consequently have law $\nu^{\otimes n}$.

Proof. Tonelli's theorem gives

$$\mathbb{P}(X_i \in B, U_i < \pi(X_i)) = \int_B \int_0^1 \mathbf{1}\{u < \pi(x)\} du d\mu(x) = \alpha(B).$$

The rejection probability is $1 - Z$. Multiplying these one-event identities over the independent generated pairs proves (5). There are $\binom{N}{n}$ disjoint retained patterns; the same product law for the ordered locations occurs for each. Summing proves (6) on measurable rectangles. Finite measures agreeing on such rectangles agree on the product sigma algebra, so the statement holds for every measurable tuple event. Independence of N then gives the mixture by the law of total probability. Division by its count mass is justified only when that mass is positive. $\square$

These are submeasure identities even for impossible patterns. In particular, they include $Z = 1$ and $n = 0$, using empty products and $0^0 = 1$. The empty tuple has its unique probability law; impossible counts are not assigned a conditional distribution by division by zero. An arbitrarily location-dependent choice of N is outside the independence assumption.

Proposition 3.2 (Poisson generation and its Laplace functional). *If $N \sim \text{Poi}(\kappa)$ independently, with $\kappa \geq 0$, then $C_N \sim \text{Poi}(\kappa Z)$ and, conditional on a possible count n , the retained tuple has law $\nu^{\otimes n}$. For every nonnegative measurable real function h ,*

$$\mathbb{E} \exp \left(- \sum_{i \text{ retained}} h(X_i) \right) = \exp \left(\kappa \int \pi(x) (e^{-h(x)} - 1) d\mu(x) \right). \quad (7)$$

Proof. For fixed n , set $N = n + r$ in the mixture of (6). The scalar mass is

$$\sum_{r=0}^{\infty} e^{-\kappa} \frac{\kappa^{n+r}}{(n+r)!} \binom{n+r}{n} Z^n (1 - Z)^r = e^{-\kappa Z} \frac{(\kappa Z)^n}{n!}.$$

The tuple factor does not depend on r . This proves the count and conditional laws, also at $\kappa = 0$. Alternatively a single generated pair contributes a Laplace factor

$$M = \int (1 - \pi + \pi e^{-h}) d\mu = 1 + \int \pi(e^{-h} - 1) d\mu \in [0, 1].$$

For N pairs the expectation is M^N . Summing its Poisson series yields $e^{\kappa(M-1)}$, which is (7). All integrands are bounded or nonnegative, so the integrations and series are justified. $\square$

Repeated locations, if the underlying measure has atoms, are counted with multiplicity. For a nonnormalized geometry of total volume V , taking $\mu = \text{vol}/V$ and generated intensity κV instead produces the retained intensity measure $\kappa \pi \text{ vol}$. This identity does not make V identifiable from a fixed retained order.

Proposition 3.3 (Stopping after the first n retained events). *In the infinite independent stream, the prefix ending at the n th retained event is almost surely finite for every $n \geq 1$. Its ordered retained tuple has law $\nu^{\otimes n}$. For $n = 0$, the prefix and tuple are empty.*

Proof. After any fixed index, the probability that the next m trials all fail is $(1 - Z)^m$, which tends to zero. By continuity of probability on decreasing events, having no later success has probability zero. If there were only finitely many successes, some integer index would be after the last one. A countable union of these null events is null. Thus every required success occurs in a finite prefix.

For $n \geq 1$ and a terminal prefix of length $N \geq n$, its last event is accepted and exactly $n - 1$ of the preceding $N - 1$ events are accepted. Applying (5) to these disjoint patterns gives

$$\mathbb{P}(T_n = N, V_n \in B) = \binom{N-1}{n-1} Z^n (1-Z)^{N-n} \nu^{\otimes n}(B).$$

These terminal events are disjoint and exhaust probability one by the first paragraph. Summing first with B the whole tuple space shows their scalar masses sum to one, and then summing for arbitrary B proves the result. At $n = 0$ the assertion is the empty-product identity. $\square$

The accepted formal proof sums a finite set of terminal patterns instead of simplifying its cardinality to $\binom{N-1}{n-1}$. The displayed combinatorial simplification is elementary prose packaging; the actual stopped-tuple law is formally verified. The stopping argument concerns a prescribed retained count, not a general data-dependent stopping rule.

4 A finite report for physical volume

The following transformation separates detector uncertainty from sampling uncertainty. Here $R \geq 1$ may exceed the range needed in the joint rate. Define, for $x \in [0, 1]$,

$$L_R(x) = \frac{x}{R - (R-1)x}, \quad U_R(x) = \frac{Rx}{1 + (R-1)x}. \quad (8)$$

Proposition 4.1 (Bounded selection). *For a probability measure μ , a measurable target set of probability v , and a detector $\pi \in [1/R, 1]$, let θ be its retained probability. Then*

$$L_R(\theta) \leq v \leq U_R(\theta), \quad |\theta - v| \leq b_R. \quad (9)$$

Both transforms are increasing on $[0, 1]$.

Proof. Put $s = \int_A \pi d\mu$ and $t = \int_{A^c} \pi d\mu$. Then $v/R \leq s \leq v$ and $(1-v)/R \leq t \leq 1-v$. Since $s+t > 0$ and $\theta = s/(s+t)$, these inequalities imply

$$\frac{v}{v + R(1-v)} \leq \theta \leq \frac{Rv}{Rv + 1 - v}.$$

Solving the two inequalities for v proves the bounds in (9), including $v = 0, 1$. All relevant denominators are positive. For $x \leq y$, the differences of the transforms are

$$L_R(y) - L_R(x) = \frac{R(y-x)}{[R - (R-1)x][R - (R-1)y]},$$

$$U_R(y) - U_R(x) = \frac{R(y-x)}{[1 + (R-1)x][1 + (R-1)y]},$$

which prove monotonicity. Finally, direct common-denominator calculations give

$$\begin{aligned} b_R - (x - L_R(x)) &= \frac{(R-1)[R(1-x)^2 + x^2]}{(R+1)[R - (R-1)x]} \geq 0, \\ b_R - (U_R(x) - x) &= \frac{(R-1)[(1-x)^2 + Rx^2]}{(R+1)[1 + (R-1)x]} \geq 0. \end{aligned}$$

Substitute $x = \theta$ and use the established bounds for v . $\square$

These binary transforms appear explicitly in Aronow and Lee [1, p. 237]. For observed zero-one values with k ones, a normalized weighted mean with weights in $[1, R]$ has minimum $L_R(k/n)$ and maximum $U_R(k/n)$: assign the smallest weights to the ones and the largest to the zeros for the minimum, and reverse them for the maximum. The analogous probability bounds are sharp when no further restrictions are imposed, by constant detectors on A and A^c . Neither fact proves sharpness of the identified set inside $\mathcal{K}$ conditional on the full order. Moreover, plug-in bounds alone are not a confidence interval. With $n = 1$, $R = 1$ and the flat density, the plug-in interval is $\{0\}$ or $\{1\}$, and has zero coverage for the true value $1/4$.

Theorem 4.2 (Finite confidence from valid endpoint tables). *Let $0 < \delta < 1$ and choose endpoints $0 \leq l(k) \leq u(k) \leq 1$ for every $k \in \{0, \dots, n\}$. Suppose, with $X_p \sim \text{Bin}(n, p)$, that*

$$\begin{aligned} l(k) = 0 \quad \text{or} \quad \mathbb{P}(X_{l(k)} \geq k) &\leq \delta/2, \\ u(k) = 1 \quad \text{or} \quad \mathbb{P}(X_{u(k)} \leq k) &\leq \delta/2. \end{aligned} \tag{10}$$

If $K \sim \text{Bin}(n, \theta)$, then

$$\mathbb{P}\{l(K) \leq \theta \leq u(K)\} \geq 1 - \delta.$$

In the geometric model, a physical-volume report with the same coverage is

$$I(K) = [L_R(l(K)), U_R(u(K))] \cap [1/8, 3/8]. \tag{11}$$

At $n = 0$, take $[l(0), u(0)] = [0, 1]$, giving $I(0) = [1/8, 3/8]$. The coverage holds in each conditional retained-count experiment of Section 3, in the first- n experiment, and after mixing over retained counts when the same failure budget is used at each count.

Proof. Couple $X_p = \sum_{i=1}^n \mathbf{1}\{U_i < p\}$ with independent uniform U_i . This count increases with p , so its upper tails increase and its lower tails decrease with p . Under the true θ , a failure $l(k) > \theta$ implies $\mathbb{P}(X_\theta \geq k) \leq \delta/2$ by (10) and monotonicity. For any integer-valued random variable X , the event $\{\mathbb{P}(X \geq X_{\text{obs}}) \leq c\}$ has probability at most c : if there is a qualifying integer in its finite support, let k_0 be the smallest; all qualifying outcomes are in $\{X \geq k_0\}$, whose mass is at most c . If none qualifies the event is empty. Applying this with $c = \delta/2$ bounds the lower-endpoint failure. The largest qualifying integer and a lower tail give the same bound for $u(k) < \theta$. The union bound proves the first assertion. No monotonicity of the endpoint table as a function of k is required.

On the successful binomial event, Proposition 4.1 and monotonicity imply $L_R(l(K)) \leq v_\rho \leq U_R(u(K))$. The deterministic range in Proposition 2.1 permits intersection without losing that event. An empty intersection stays empty; it has no target coverage. The zero-sample statement is immediate. The tuple laws already proved transfer this argument to each conditional or stopped experiment. Finally, a mixture of failure probabilities each at most δ is at most δ . $\square$

Proposition 4.3 (Integer endpoint certificates). *For $p = m/d$ with integers $0 \leq m \leq d$ and $d > 0$, binomial probabilities can be certified exactly using*

$$\mathbb{P}(X_p = j) = \frac{t_j}{d^n}, \quad t_j = \binom{n}{j} m^j (d - m)^{n-j}.$$

For $0 < m < d$ and $j < n$, the recurrence

$$t_{j+1} = \frac{t_j (n - j) m}{(j + 1)(d - m)} \tag{12}$$

has exact integer division. Rational tail comparisons in (10) therefore reduce to integer comparisons.

Proof. Substitution in the binomial mass formula gives the common denominator d^n . The identity $(j + 1) \binom{n}{j+1} = (n - j) \binom{n}{j}$ proves (12) by multiplication; its denominator is positive and the quotient equals the integer t_{j+1} . Sum these numerators over the desired tail and cross-multiply positive denominators against the rational budget. At $m = 0$ or $m = d$, use the deterministic count at 0 or n instead of dividing by zero. The empty-sample mass is one. $\square$

The implementation proposes Clopper–Pearson endpoints using floating point beta quantiles, rounds them outward to denominator 2^{16} , then checks the rational tail inequalities exactly. Proposal precision is not the certificate: a failed check triggers an outward adjustment or a full-range fallback. Both preserve the stated coverage when the final table passes. The recorded pilot checks every count in its tables; its adjacent inward checks additionally bound the outward grid error for those tables. This is not a guarantee on every possible proposal/fallback path.

For comparison we also use $[\max(0, K/n - r_n), \min(1, K/n + r_n)]$ with a rational $r_n \geq \sqrt{2/n}$. Hoeffding’s bound is at most $2e^{-4} < 1/20$: indeed $e^4 \geq \sum_{j=0}^5 4^j/j! > 40$. Transforming these endpoints gives a valid analytic physical report. The implementation chooses the rational radius by an integer square-root calculation. At $n = 0$ it uses the no-data interval. Python execution and floating point proposal semantics are tested software, not compiled Lean theorems.

5 Proof of the joint accuracy theorem

We first give the upper argument, then prove two lower obstructions for the actual marked observation.

5.1 Uniform upper bound

By Proposition 4.1, $|\theta - v_\rho| \leq b_R$. The count fraction satisfies $0 \leq K/n \leq 1$ and $v_\rho \in [0, 1]$. If $a_n + b_R \geq 1$, error strictly greater than $2s = 2$ is impossible. Otherwise, error greater than $2(a_n + b_R)$ forces $|K/n - \theta| > 2a_n + b_R \geq 2a_n$. Hoeffding’s inequality therefore gives

$$\mathbb{P}\{|K/n - v_\rho| > 2s(n, R)\} \leq 2 \exp(-2n(2a_n)^2) = 2e^{-8} \leq 1/20.$$

This proves (3); the same argument works for any $R \geq 1$. The restriction $R \leq 2$ is used below in the geometric detector alternative.

5.2 Testing, geometry and actual likelihoods

Lemma 5.1 (Testing with an independent seed). *Let P, Q be probability laws on a finite observation space, and suppose $\text{TV}(P, Q) \leq \eta$. For any common independent seed law ξ , measurable real estimator T and targets $v_0 < v_1$, if $2r < v_1 - v_0$, then*

$$\max\{(P \otimes \xi)(|T - v_0| > r), (Q \otimes \xi)(|T - v_1| > r)\} \geq (1 - \eta)/2.$$

In particular, identical laws give the lower bound $1/2$.

Proof. Let $B_i = \{|T - v_i| \leq r\}$. These measurable sets are disjoint: membership in both would give $v_1 - v_0 \leq 2r$. For any measurable set B in the observation–seed product, its section probabilities $h(y) = \xi\{\omega : (y, \omega) \in B\}$ lie in $[0, 1]$. The finite signed-mass formula gives

$$|(P \otimes \xi)(B) - (Q \otimes \xi)(B)| = \left| \sum_y (P\{y\} - Q\{y\})h(y) \right| \leq \text{TV}(P, Q).$$

Thus $(P \otimes \xi)(B_0) + (Q \otimes \xi)(B_1) \leq (Q \otimes \xi)(B_0 \cup B_1) + \eta \leq 1 + \eta$. The complementary failure probabilities sum to at least $1 - \eta$, which proves the claim. The identical-law argument also works on an arbitrary measurable observation space. $\square$

Lemma 5.2 (An admitted polynomial family). *For $0 \leq \varepsilon \leq 1/2$, define*

$$f(t) = 2t - 1, \quad \rho_\varepsilon(u, v) = 1 + \varepsilon f(u)f(v).$$

Then $\rho_\varepsilon \in \mathcal{K}$ and

$$v_{\rho_\varepsilon} = 1/4 + \varepsilon/16. \tag{13}$$

Under n unthinned flat coordinate samples, the likelihood of the ρ_ε sample is $L = \prod_i \rho_\varepsilon(X_i)$, with

$$\mathbb{E}_0 L = 1, \quad \mathbb{E}_0 (L - 1)^2 = (1 + \varepsilon^2/9)^n - 1. \tag{14}$$

Proof. The polynomial is smooth. On $[0, 1]$, $|f| \leq 1$, giving $1 - \varepsilon \leq \rho_\varepsilon \leq 1 + \varepsilon$. For $x = (u, v)$ and $y = (u', v')$ in D , adding and subtracting $f(u')f(v)$ yields

$$|\rho_\varepsilon(x) - \rho_\varepsilon(y)| \leq 2\varepsilon(|u - u'| + |v - v'|) \leq 4\varepsilon\|x - y\|_2 \leq 2\|x - y\|_2.$$

Since $\int_0^1 f = 0$, both coordinate marginals equal one. Also $\int_0^{1/2} f = -1/4$, proving (13). The flat law is uniform on D . Multiplication of the single-point density identities gives the actual product likelihood on coordinate space; it is nonnegative on that support and integrates to one. Because $\int_0^1 f^2 = 1/3$, the single-point second moment is $1 + \varepsilon^2/9$. Independence gives its n th power for $\mathbb{E}_0 L^2$. Subtracting $2\mathbb{E}_0 L - 1 = 1$ proves (14). $\square$

Lemma 5.3 (Sampling obstruction). *For every $n \geq 1$, $R \geq 1$ and estimator/seed in Theorem 2.2, there is an admitted model with unit detector such that*

$$(P_{\rho,1}^{(n)} \otimes \xi)(|T - v_\rho| > a_n/128) \geq 3/8.$$

Proof. Compare $\rho_0 = 1$ with ρ_ε for $\varepsilon = 1/(2\sqrt{n}) \leq 1/2$, using $\pi = 1$ for both. This detector is allowed for every $R \geq 1$. Put $x = 1/(36n)$. Bernoulli's inequality gives $(1 - x)^n \geq 1 - nx = 35/36$; it follows by induction from $(1 - x)^{j+1} \geq (1 - jx)(1 - x) \geq 1 - (j + 1)x$ whenever needed here. Moreover $(1 + x)^n(1 - x)^n = (1 - x^2)^n \leq 1$. Hence

$$\mathbb{E}_0 (L - 1)^2 = (1 + 1/(36n))^n - 1 \leq 1/35.$$

For coordinate laws with density L relative to the flat law, their total variation is $\frac{1}{2}\mathbb{E}_0|L - 1|$; this follows by splitting the zero-integral function $L - 1$ into positive and negative parts. Cauchy–Schwarz therefore bounds it by $1/(2\sqrt{35}) \leq 1/4$.

Mapping coordinates to the full marked order cannot increase total variation: the inverse image of an observation event is a coordinate event. The two full marked-order laws thus have total variation at most $1/4$, regardless of the other relations present in the observation. Their targets differ by $\varepsilon/16 = a_n/32$. With $r = a_n/128$, the strict separation $2r < a_n/32$ holds. Lemma 5.1 now gives the asserted lower bound $(1 - 1/4)/2 = 3/8$. $\square$

5.3 Exact detector confounding

Lemma 5.4 (Calibration obstruction). *For $1 < R \leq 2$, $n \geq 0$ and any eligible estimator/seed, there is an admitted model satisfying*

$$(P_{\rho,\pi}^{(n)} \otimes \xi)(|T - v_\rho| > b_R/64) \geq 1/2.$$

The two alternatives used in the proof have the same unnormalized accepted-location measure. Their finite generated-count and stopped accepted-tuple experiments also have the same laws.

Proof. Set $\varepsilon = b_R \in (0, 1/3]$. Compare the flat density with ρ_ε from Lemma 5.2. Define the detectors globally by

$$\pi_0(x) = 1 - \varepsilon, \quad \pi_\varepsilon(x) = \frac{1 - \varepsilon}{\max\{1 - \varepsilon, \min\{1 + \varepsilon, \rho_\varepsilon(x)\}\}}.$$

The denominator is continuous and at least $1 - \varepsilon > 0$. Thus π_ε is measurable and everywhere lies between $(1 - \varepsilon)/(1 + \varepsilon) = 1/R$ and 1. Also $1 - \varepsilon \geq 1/R$, so π_0 is admitted. The clamping matters outside D ; an unclamped reciprocal need not be a valid global detector.

On D the denominator equals ρ_ε , so $\rho_\varepsilon \pi_\varepsilon = 1 - \varepsilon = \rho_0 \pi_0$. Both unnormalized accepted measures equal $(1 - \varepsilon) \text{Leb}|_D$, both acceptance masses equal $1 - \varepsilon$, and both retained laws are uniform on D . Their full marked-order laws are therefore identical for every n . Their target volumes differ by $\varepsilon/16$, which is strictly larger than $2(\varepsilon/64)$. The identical-law case of Lemma 5.1 proves the claim.

Equality of the accepted submeasures gives equality of Z and ν . Equations (5)–(6) then give equality even if the generated count and acceptance pattern are retained, provided the same generated-count law is used. The mixed, Poisson and stopped identities give their respective equalities as well. Rejected locations are not part of these equal-law records. $\square$

5.4 Combining the obstructions and treating zero samples

Completion of Theorem 2.2. Only the lower bound remains. If $a_n \geq b_R$, then $s(n, R) \leq 2a_n$ and $s(n, R)/256 \leq a_n/128$. Lemma 5.3 supplies failure at least $3/8$, hence at least $1/4$, at the smaller radius in (4). If $b_R > a_n$, then $R > 1$, $s(n, R) \leq 2b_R$, and $s(n, R)/256 \leq b_R/128 < b_R/64$. Lemma 5.4 supplies failure at least $1/2$ at the larger radius, and therefore at least $1/4$ at the stated radius. These cases exhaust all $n \geq 1$ and $1 \leq R \leq 2$, including $R = 1$. $\square$

Proposition 5.5 (Zero retained samples). *For $n = 0$, the constant estimator $1/4$ has error at most $1/8$ over $\mathcal{K}$. For every $R \geq 1$ and every measurable randomized estimator of the empty-sample marked observation, some admitted model satisfies*

$$(P_{\rho,\pi}^{(0)} \otimes \xi)(|T - v_\rho| > 1/128) \geq 1/2.$$

Proof. The upper statement follows from $v_\rho \in [1/8, 3/8]$. For the lower statement, compare the flat density and $\rho_{1/2}$, both with unit detector. There are no random sampled points. The observation still includes the fixed anchor relations, but these are identical in both models, so its law is the same Dirac mass. The targets differ by $1/32 > 2/128$. Apply the identical-law testing argument with $r = 1/128$ and the arbitrary independent seed. $\square$

6 Two preserved computational studies

The following studies were frozen and executed before preparation of this manuscript. Their data are preserved unchanged. They ask different questions and do not form one pooled experiment. All reported widths are dimensionless volume-fraction widths. An expected width averages over a specified count law; a realized width is the width at one observed count. Coverage means probability of containing the true target and is distinct from either width.

6.1 Original generic-probability pilot

The pilot used 300 combinations of retained count, target probability, detector ratio and detector pattern, and four methods, producing 1,200 rows. Counts were $n \in \{0, 64, 256, 1024\}$. Each configuration used 2,000 Monte Carlo counts as well as a numerical sum over the complete binomial law. At $n = 0, 64, 600$ method rows also received exact rational coverage checks; 450 belonged to the three valid methods and 150 to the deliberately misspecified method that ignored detection bias. The generic target grid includes probabilities outside $[1/8, 3/8]$; it is not a grid confined to our fixed geometric interval.

The valid methods were the transformed certified binomial interval, the transformed analytic interval, and the full $[0, 1]$ no-data interval. The fourth method estimated the detected probability as if it were the physical probability. The 1,348 distinct count intervals passed their exact endpoint checks, with 2,688 adjacent inward endpoint rejections and no adjustments or fallbacks. Minimum complete-law numerical coverage over valid methods was 0.951071; the minimum exact rational coverage in the checked small-count subset was 0.966807. The bias-ignoring method fell to approximately 1.20×10^{-19} .

R	Certified binomial	Analytic
1	0.06214	0.08839
1.1	0.10938	0.13539
1.25	0.17195	0.19746
1.5	0.25874	0.28320
2	0.38717	0.40939

Table 1: Original pilot: worst expected widths on its prescribed generic-probability grid at $n = 1024$. The no-data width is one.

Table 1 reports the worst *expected* width over this pilot’s prescribed target and detector patterns at $n = 1024$. Its frozen width threshold of 0.20 passed for all tested cells with $R \leq 1.25$. This was a bounded feasibility result, not a claim over untested targets or a field detector. Execution took 277.11 seconds, with 4,774,983 bytes of Python-traced peak memory. That metric does not include all native allocations or measure total process resident memory.

The pilot separately generated four actual geometric configurations, each with 200 trials of $n = 1024$ retained events: flat density or $\rho_{1/4}$, with constant or compensating detection. Their

targets were respectively $1/4$ and $17/64$. Recorded coverage proportions were 0.935, 0.955, 0.965 and 1.000 in the preserved artifact order. The first is 187 successes out of 200; it has not been replaced or hidden because it lies below 0.95. Its exact binomial confidence interval for the repeated-trial success probability is approximately $[0.8914, 0.9650]$; the complete-law coverage for the corresponding flat case is about 0.952905. Thus this Monte Carlo fluctuation does not contradict the finite guarantee. The compensated polynomial case has $R = 5/3$ and mean width about 0.24188.

6.2 Frozen geometric finite comparison

The later study used the known geometric range $[1/8, 3/8]$ for every method. It compared four reports from the same (n, K, R) : transformed certified binomial, transformed Hoeffding, the joint-rate interval centered at K/n with radius $2s(n, R)$, and the fixed known range. Every interval was intersected with the known range. An empty intersection had zero width and no coverage; it was not replaced by a fallback. All methods returned the known range at $n = 0$.

The formula grid used $n = q^2$ for $q = 1, 2, 4, 8, 16, 32, 64$ and

$$R \in \{1, 1025/1024, 65/64, 17/16, 9/8, 5/4, 3/2, 2\},$$

augmented by admitted crossover values with $b_R \in \{1/(2q), 1/q, \min(2/q, 1/3)\}$ and $b_R \leq 1/3$. There were 71 rate cells. Report calculations included every count for $n = 0, 1, 4, 16, 64$ and seven selected counts $0, 1, n/4, n/2, 3n/4, n-1, n$ for $n = 256, 1024$. All 104 distinct binomial endpoint reports passed exact tail checks with no fallback.

For complete-law coverage, the study used ρ_ε with $\varepsilon \in \{0, 1/4, 1/2\}$ and uniform, inside-low or inside-high piecewise detection in $[1/R, 1]$. For $n \leq 64$, all 1,656 exact rational coverage checks passed; their minimum was approximately 0.96288634. Expected widths were enclosed by outward rational display intervals of width at most 10^{-12} , and empty-report probabilities were computed exactly. Complete-law coverage was not calculated at $n = 256$ or 1024 in this study.

Across 1,108 paired realized-report cases, the joint-rate interval was wider than the certified binomial report in 732 and equal in 376; it was never narrower on that grid. Table 2 shows three selected *realized* widths at $n = 1024$, $K = 256$. This is evidence for retaining the exact-binomial practical report, not a proof of universal interval dominance or shortest possible intervals.

R	Certified binomial	Hoeffding	Joint rate	Known range
1	0.053970	0.088388	0.125000	0.250000
1.25	0.137225	0.170836	0.250000	0.250000
2	0.249039	0.250000	0.250000	0.250000

Table 2: Separate geometric study: realized widths at $n = 1024$, $K = 256$, after intersection with the known range.

The formula checks included eight polynomial parameter values, product divergence through $n = 4096$, and five negative controls: an incorrect target denominator, an incorrect second moment, an invalid strict-gap boundary, an unclamped ambient detector and a shrunken binomial interval. No Monte Carlo samples, numerical quadrature or random seeds were used in this second study. Execution took 199.33 seconds with 59,031,222 bytes of Python-traced peak memory. A separate artifact audit took about 3.15 seconds and checked the saved cells, including 261 binomial-mass recurrences. The first attempt stopped because SciPy was unavailable; its partial output remains preserved. The successful retry used the same frozen protocol and runner after restoring the required dependency. It did not alter the grid or select a favorable resampling outcome.

7 Source comparisons and verification scope

Aronow and Lee’s journal article and two-page supplement were read in full, including the article’s Proposition 1 and the supplement’s Lemma 1 and Propositions 2–3. Their weighted-mean bounds, binary formulas and asymptotic convergence arguments receive attribution here. They do not themselves supply this paper’s finite binomial confidence calibration or the full marked-order geometric lower construction. In the reviewed comparison, their binary transform is algebraically identical to the transform already applied to the calibrated binomial benchmark. Adding another numerical column with those same endpoints would duplicate the existing method.

The related partial-identification comparisons use the editions actually inspected: Imbens and Manski’s 2003 CeMMAP working paper, Stoye’s 17 December 2008 preprint, and Tudball and colleagues’ fourth arXiv version together with the journal and supplementary proof comparison. Their confidence constructions and sample constraints have their own regularity and asymptotic assumptions. We have not imported such assumptions to replace our finite count argument, nor inferred that every possible partial-identification method is covered by this comparison. An uninspected journal edition is not claimed textually identical to its reviewed preprint. Purchased source PDFs are kept private and are not redistributed with our source package.

The accepted Lean development includes exact types and axiom reports for 797 exports in its original full audit. That total covers the entire accepted dependency collection, not 797 new theorems of this manuscript. The authoritative run is

`space-volumerate-acceptance-20261010T023105Z-5d08cf.`

It ran on Google Cloud with Lean 4.30.0 and the pinned mathlib revision recorded in the source archive. It completed 3,097 root build jobs with zero warnings. Its reports use only the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The 198 captured source files, dependency identities, raw logs, receipt and task-owned instance shutdown evidence are retained. Proof commit:

`354fa1f8dc8746328077041ef79d3621837f99e7.`

The accompanying `formal-map.json` associates each labeled mathematical claim with the exact accepted exports and source hashes. It also identifies elementary prose packaging, such as the determinant calculation and terminal-pattern cardinality, rather than representing every sentence as a separate kernel theorem. Principal entries are:

Manuscript result	Principal Lean module(s)
Finite/mixed thinning	<code>ThinningCount</code> , <code>ThinningMixture</code>
Poisson and stopped laws	<code>ThinningPoisson</code> , <code>ThinningStoppedLaw</code>
Finite confidence and certificates	<code>BinomialCalibration</code> , <code>BinomialArithmetic</code>
Physical report transfer	<code>BinomialGeometry</code> , <code>ThinningMixedCoverage</code>
Geometric alternatives	<code>ConfoundingGeometry</code> , <code>ConfoundingMarked</code>
Sampling lower bound	<code>VolumeRateLikelihood</code> , <code>VolumeRateSampling</code>
Uniform upper/lower theorem	<code>VolumeRateUpper</code> , <code>VolumeRateJoint</code>
Zero samples	<code>VolumeRateEmpty</code>

Preparation reuses this immutable acceptance; it does not rerun Lean or change any accepted proof. The source verifier checks the full captured archive against the original proof commit and the current mapped modules against that archive. Later repository integration files can contain additional projects without changing the original acceptance evidence. The Python reports and mathematical source comparison are separately audited evidence, not kernel verification of Python,

SciPy, the operating system or an actual detector. Formal correctness also does not establish that the model is a correct description of a physical instrument.

The reproducibility package supplies the manuscript source and build instructions, exact formal map, original proof snapshot and acceptance logs, table extraction, and frozen computational evidence. Its manifest records file hashes. The manuscript is licensed CC BY 4.0; source-code licenses and the original repository license are preserved. This is an unpublished preparation with no newly assigned manuscript DOI.

8 Implications and limits

With perfectly calibrated uniform detection ($R = 1$), the theorem gives the familiar $n^{-1/2}$ scale for this scalar target. For fixed $R > 1$, the calibration term remains positive as n grows. Within the stated model, using more marked-order relations cannot force the worst-case volume error to vanish: the compensated geometries have exactly the same retained law. This is an information limit of the observation experiment, not a failure of one estimator.

The sampling and calibration terms become comparable at $n^{-1/2} \asymp (R - 1)/(R + 1)$. To keep a sampling-order accuracy as n increases, the ratio uncertainty must shrink at that order as well. This gives a statistical way to distinguish the benefit of more retained events from the benefit of better detector knowledge. It does not choose a monetary optimum, specify a calibration instrument, or measure a sensor’s achievable R . If R were inferred from finite calibration data, its uncertainty would need an additional coverage argument and failure-budget allocation.

Physical use would require evidence for the causal observation map, the sampling and independent detection assumptions, a justified bound on the detector ratio, stable anchor identification and any conversion from normalized to absolute units. None is demonstrated by the synthetic studies. Dependence between detections, geometry-dependent stopping, measurement error in causal relations or sample-selected anchors are outside the present theorem. The results do not establish reconstruction in higher dimensions, curvature estimates, operational resource gains or a new quantum inference method.

The practical outcome is an inspectable finite report together with a proved explanation of its calibration limit. Open-source, decentralized review and downstream use can test and improve these results; anticipated uptake is not treated as validation. New applications should first show that their observation and calibration assumptions fit the experiment, then compare the report with an established method using the same data and coverage requirements.

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